PARTNERSHIP MEMO

United States–Germany Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.

Executive Decision Summary

BUILD TOGETHER

Battery Materials & Cells: Synthetic graphite (anode-grade) (Germany->US); Natural graphite (joint); Other accumulators (joint). Electric Vehicles: Electric & plug-in cars (Germany->US). Nuclear Fuel & Reactor Systems: Nuclear extraction (joint). Permanent Magnets & Drive Systems: Rare-earth oxides & compounds (joint); Ferrite & other magnets (Germany->US); Electric motors (Germany->US); Motor & generator parts (Germany->US). Semiconductors & Power Electronics: Power electronics components (Germany->US).

TRADE AND STANDARDS

Recognise trusted partner supply of electric vehicles refining (rare-earth metals, rare-earth oxides & compounds); electric & plug-in cars; ferrite & other magnets in procurement and content rules. Bind market access for synthetic graphite (anode-grade). Keep uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores); nuclear special materials; transformers open to global sourcing.

INSTITUTIONS

Upgrade US-EU critical minerals MOU and action plan (EU-level, applies to Germany): sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

STAY OUT

Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity. Do not co-subsidise new Automatic controllers capacity. Do not co-subsidise new Electric buses, trucks & tractors capacity.

Why this partnership matters.

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 45 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 7 direct competition and 67 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: US Demand & Market Anchor (primary), Coordination / Bridge Hub (primary), Capital & Investment Anchor (primary), Scale Manufacturing & Integration Platform (secondary), Specialized Technology & Manufacturing Anchor (secondary); Germany Capital & Investment Anchor (primary), Scale Manufacturing & Integration Platform (primary), Demand & Market Anchor (primary), Coordination / Bridge Hub (secondary), Specialized Technology & Manufacturing Anchor (secondary).[1]

The central bilateral bargain.

- *Inference.* Germany supplies synthetic graphite (anode-grade), electric & plug-in cars, ferrite & other magnets, electric motors where the US is short; the US supplies battery recycling feed, refrigerants, sensors & controller parts, ac generators where Germany is short; the two share exposure to the same coercive supplier in rare-earth oxides & compounds, natural graphite, other accumulators, li-ion cells & packs, rare-earth permanent

magnets, copper smelting & refining, where neither can supply the other and the answer is joint demand, finance and third-country capacity.

- *Inference.* From the AIB roles, the US brings demand and offtake, coordination across allied chains, capital; Germany brings capital, scale manufacturing, demand and offtake. Public finance runs through DOE LPO, DoW OSC, DFC, EXIM on the US side and KfW, KfW IPEX, Hermes, UFK, DEG on the Germany side. The trade is market access and demand certainty for trusted capacity.

The 5 highest-priority joint bets (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

PRIORITY	SECTOR	NODES IN THE BET (DIRECTION)	STRATEGIC PROBLEM	US CONTRIBUTION	GERMANY CONTRIBUTION	MAIN INSTRUMENT (LEAD NODE)	TIMING
1	Battery Materials & Cells	Synthetic graphite (anode-grade) (Germany->US); Natural graphite (joint); Other accumulators (joint)	geopolitical chokepoint	Synthetic graphite (anode-grade): demand and long-term offtake; Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply	Synthetic graphite (anode-grade): trusted capacity (6% of world supply); Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months
2	Electric Vehicles	Electric & plug-in cars (Germany->US)	geopolitical chokepoint	Electric & plug-in cars: demand and long-term offtake	Electric & plug-in cars: trusted capacity (26% of world supply)	anchor offtake	12-24 months
3	Nuclear Fuel & Reactor Systems	Nuclear extraction (joint)	geopolitical chokepoint	Nuclear extraction: its demand for pooled offtake, and finance for non-Chinese supply	Nuclear extraction: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months
4	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds (joint); Ferrite & other magnets (Germany->US); Electric motors (Germany->US); Motor & generator parts (Germany->US)	geopolitical chokepoint	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Ferrite & other magnets: demand and long-term offtake	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Ferrite & other magnets: trusted capacity (6% of world supply)	anchor offtake	12-24 months
5	Semiconductors & Power Electronics	Power electronics components (Germany->US)	trusted partner capacity constraint	Power electronics components: demand and long-term offtake	Power electronics components: trusted capacity (7% of world supply)	anchor offtake	12-24 months

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (oil extraction); they are handled in the trade agenda, not as compacts.

Do not prioritise (Section 10):

- *Recommendation.* Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity: both are strong suppliers (US 7%, Germany 8%). Trusted supply is adequate: this is commercial competition.

- *Recommendation.* Do not co-subsidise new Automatic controllers capacity: both are strong suppliers (US 16%, Germany 12%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* Do not co-subsidise new Electric buses, trucks & tractors capacity: both are strong suppliers (US 15%, Germany 11%). More trusted capacity is needed, so agree who specialises in what rather than racing.

Existing mechanisms and the MEA question (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (USDE-M3)**. Gaps: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-EV, SC-SEMIS, SC-GRID; USDE-M3 has no published project pipeline.
- *Recommendation.* Do not sign a new energy MOU. Give USDE-M3 a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

Alliance and plurilateral architecture

- Germany's memberships in OSI's economic-security agreements table: G7, MSP, NATO.[1]
- Germany and the USA are both original MSP partners and automatic members of the MSP Forum launched in 2024.[2]
- TTC (created 2021) had its future questioned under the 2025 administration; no formal termination or revival was found, so it is treated as dormant and superseded by the 2025 framework and 2026 critical minerals MOU.[3]
- G7 leaders (including the USA and Germany) launched the Action Plan at Kananaskis on 17 Jun 2025 to diversify production, encourage investment and local value creation; interested members support the Minerals Security Partnership; no named German-US projects.[4]

Trade architecture and frictions

- In the OSI energy basket, US imported US$27.8bn from Germany and Germany imported US$26.8bn from US (bilateral flows 2024, summed over the technology x stage cells).[1]
- Parliament approved the main regulation on 16 Jun 2026 (440-151), governments adopted it on 25 Jun and EU tariff cuts took effect 1 Jul 2026; Parliament's position carried suspension and sunset clauses (sunset 31 Mar 2028).[5]
- Parliament suspended work on approving the deal on 21 Jan 2026 over US Greenland demands and tariff threats and unfroze it in early Feb 2026; the freeze is lifted, so status shown is historical.[6]
- Supreme Court held on 20 Feb 2026 that IEEPA does not authorise tariffs; a Section 122 surcharge (reported at 10%, then 15%) ran from 24 Feb for 150 days and could exceed the Turnberry 15% ceiling where MFN is zero (Bruegel); it expired 24 Jul 2026; sources conflict on the rate.[7]
- On 31 Jul 2026 the Commission removed the 6 Aug expiry and extended indefinitely the suspension of EUR 93bn of countermeasures (Regulation 2025/1564), under continuous review with the option to reactivate; page seen only as a search result.[8]
- GM terminated its long-term magnet supply agreement with e-VAC (reported 22 Oct 2026); the parties are reportedly in arbitration and the dispute dates from April 2026; one German source dates the notice to July 2026; reasons not stated by the parties.[9]

Industrial and energy agreements

- Germany and the US signed an energy and climate partnership in May 2022 with work on LNG, hydrogen and offshore wind; no 2025-26 bilateral energy, hydrogen or critical-minerals agreement was found, and current activity is unconfirmed.[10]
- US applies the higher of MFN or a 15% all-in tariff on EU goods; the EU intends US$750bn of US LNG, oil and nuclear purchases through 2028 and US$40bn of AI chips, and European firms are expected to invest US$600bn more in the US; commitments mostly non-binding.[11]
- Rubio and Sefcovic signed a non-binding MOU on 24 Apr 2026 to diversify mining, processing and refining and coordinate on third-country export controls; a USTR action plan explores border-adjusted price floors and a later plurilateral club; no binding agreement or projects found.[12]

Investment vehicle, project pipeline and financing

Corporate relationships in the recommended compacts

- Siemens Energy announced a US$150m large power transformer plant in Charlotte in Feb 2024, originally for early-2026 output (24 units a year, rising to 57); a June 2025 report put US production at 2027.[13]
- e-VAC was awarded about US$112m in Section 48C tax credits for its US$500m-plus Sumter facility in April 2024.[14]
- Infineon agreed in Feb 2025 to sell its 200 mm Austin fab to SkyWater with a long-term supply agreement, reducing its own US wafer capacity (about 4,000 US staff remain).[15]
- Siemens announced more than US$10bn of added US investment in March 2025, including electrical-products plants in Fort Worth and Pomona (US$285m reported) and the Altair acquisition; the figure combines capex and an acquisition.[16]
- Siemens Energy announced US$1bn of US expansion across transformers, grid equipment and turbines, with a greenfield high-voltage switchgear plant in Mississippi (up to 300 jobs).[17]
- Mercedes plans a new Tuscaloosa model by 2027 (GLC localisation reported later); VW's Scout plant in South Carolina (about US$2bn) targets EV SUVs and pickups from late 2026-2027, while ID.4 output at Chattanooga was reported ending April 2026; sources conflict.[18]
- US-owned Tesla raised planned on-site 4680 cell capacity at Gruenheide from 8 to 18 GWh a year with about US$250m, cell output from H1 2027; the plant currently uses CATL and LG Energy Solution cells.[19]
- the US$800m figure is from a secondary source; publication date approximate.[20]
- BMW's US$1.7bn US programme includes a roughly US$700m Woodruff pack plant using Gen6 cylindrical cells from AESC (Florence SC) and iX5 production at Spartanburg, with series pack output reported for Dec 2026; publication date approximate.[21]
- US-listed Energy Fuels agreed on 23 Jun 2026 to buy 100% of Hanau-based VAC from Ara Partners (US$718m cash plus shares); close expected early 2027 subject to US and European approvals; price may adjust with the GM arbitration.[22]
- KfW's Raw Materials Fund (part of the Germany Fund) made its first two equity decisions, EUR 150m in Vulcan Energy (lithium) and EUR 50m in Arafura's Nolans rare earths project; a third is in due diligence and the target size is EUR 1.5bn; no US-linked project named; USD at about 1.16 per EUR.[23]
- German-owned VAC's Sumter plant (about US$500m, nearly 2,000 t/yr, operating since late 2025) shipped first magnets and held an opening ceremony in July 2026; about 90% of output was contracted to GM; expansion room reported up to 12,000 t.[24]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]

- *Inference.* Working channels today: US-EU framework trade arrangement and its 2026 legal basis (EU-level, applies to Germany) (finance); G7 Critical Minerals Action Plan and Minerals Security Partnership (a project pipeline). What the relationship lacks: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-EV, SC-SEMIS, SC-GRID; USDE-M3 has no published project pipeline.

2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **US · Germany**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Batteries	W/H · W/H	M/L · M/L	S/L · M/H	S/H · M/H	S/H · S/H	-/H · -/H
Coal	S/L · W/H	M/H · M/H	M/L · S/L	M/L · S/L	S/L · S/L	-/- · -/-
Electric Grid	M/H · W/H	W/H · W/H	M/L · S/L	M/L · S/L	-/L · -/L	-/- · -/-
Electric Vehicles	M/H · W/H	M/H · W/H	S/H · S/H	S/L · S/L	S/L · S/H	-/- · -/-
Gas	S/L · W/H	S/L · W/L	S/L · M/L	M/L · M/L	S/H · M/L	-/- · -/-
Geothermal	-/- · -/-	-/- · -/-	-/- · -/-	-/- · -/-	S/L · M/L	-/- · -/-
Green Hydrogen	M/- · W/-	W/- · W/-	-/- · -/-	-/- · -/-	-/L · -/H	-/- · -/-
Nuclear	W/H · W/H	W/H · M/L	M/L · M/L	M/L · S/L	S/L · M/L	-/- · -/-
Oil	S/L · W/H	S/L · M/L	M/L · M/L	M/L · S/L	S/L · S/L	-/H · -/H
Solar	M/H · W/H	W/H · W/L	S/L · M/L	S/L · S/L	S/L · S/H	-/- · -/-
Wind	S/- · W/-	M/H · W/L	S/L · M/L	S/L · S/L	S/L · S/L	-/- · -/-

Sources: 1.

Electro-industrial nodes where either country has a position or a need (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 74 of 76 nodes:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A1 in the appendix (74 rows) carries this table in full; the CSV outputs carry every column.

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for US in: Green Hydrogen assembly (0.95); Green Hydrogen components (0.93); Nuclear components (0.92); Nuclear assembly (0.90); Batteries assembly (0.87); for Germany in: Green Hydrogen assembly (0.96); Green Hydrogen extraction (0.90); Green Hydrogen components (0.89); Batteries assembly (0.87); Electric Grid refining (0.83).[1]
- Evidence quality: 40 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 4; b_supplies_a_needs 1; complementary_specialization 16; shared_vulnerability 9; capacity_constrained_trusted_supply 15; direct_competition 7; low_strategic_case 67; deployment_demand 5; not_assessed 18.[1]

Inference. Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A2 in the appendix (52 rows) carries this table in full; the CSV outputs carry every column.

- Low-priority / open-market nodes (44): wafers, solar cells, cathode materials & precursors, pv modules & generators, rare-earth metals, cobalt intermediates & chemicals, wind turbine generating sets, lithium chemicals, wind blades & hubs, accumulator parts & separators, battery materials (foil, electrolyte), manganese dioxide (emd), nickel intermediates & chemicals, lithium minerals, cobalt ores, rare-earth ores (monazite), piezo-electric quartz, pv cover glass, heat pumps & ac units, gallium, germanium & indium, grain-oriented electrical steel (goes), electrolysers, refrigeration & heat-pump compressors, power semiconductors, insulators & bushings, electricity meters, integrated circuits, transformers, switchgear & protection, hydro turbine parts, graphite electrodes, blade core wood (balsa), uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), uranium conversion & enrichment, copper ores & concentrates, overhead conductors, power cables, aluminium wire rod, direct-reduced iron (dri/hbi), pgms (iridium, ruthenium), wind towers, nuclear special materials, hydrogen, ammonia & methanol, hydro turbines.[1]
- Deployment cells are demand context, not supply relations: high deployment need in Green Hydrogen (DEU); Batteries (USA and DEU); Electric Vehicles (DEU); Solar (DEU); Gas (USA).[1]

4. Trade Agenda

- *Inference.* **What the US should export to Germany:** Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds); Batteries components (accumulator parts & separators); Oil extraction; Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite); Battery recycling feed; Batteries assembly (li-ion cells & packs, other accumulators); Solar components (insulators & bushings, power electronics components, power semiconductors); Gas extraction; Refrigerants; Sensors & controller parts; AC generators; Silver powder; Reactor parts; Semiconductor manufacturing equipment; Oil refining; Coal extraction; Wind components (ac generators, ferrite & other magnets, insulators & bushings).
- *Inference.* **What Germany should export to the US:** Synthetic graphite (anode-grade); Electric & plug-in cars; Ferrite & other magnets; Electric motors; Motor & generator parts; Power electronics components; Electric two-wheelers & works trucks; Static converters & inverters; Solar thermal water heaters; Non-oriented electrical steel (NGO); Electric furnaces & heating; Furnace parts; Nuclear reactors & enrichment machinery; Fuel assemblies & zirconium; Oil assembly (ac generators); Coal components; Coal assembly; Electric rail vehicles; Copper wire & rod.
- *Inference.* **Where tariffs would undermine allied specialisation:** synthetic-graphite-anode-grade.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** Electric Vehicles|Refining, electric-and-plug-in-cars, ferrite-and-other-magnets, electric-motors, motor-and-generator-parts.
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** none; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** uranium-fuel-upstream--aggregated, nuclear-special-materials, transformers, uranium-conversion-and-enrichment, overhead-conductors, wind-towers.

Table A3 in the appendix (25 rows) carries this table in full; the CSV outputs carry every column.

- The tariff measures in force or in dispute between the two, the baseline any binding of the nodes above works from:

- Joint statement caps combined MFN plus Section 232 at 15% for EU pharmaceuticals, semiconductors and lumber, and applies a combined 15% to EU autos and parts (conditional on the EU tabling its tariff legislation); steel and aluminium left to possible tariff-rate quotas.[26]
- Supreme Court held on 20 Feb 2026 that IEEPA does not authorise tariffs; a Section 122 surcharge (reported at 10%, then 15%) ran from 24 Feb for 150 days and could exceed the Turnberry 15% ceiling where MFN is zero (Bruegel); it expired 24 Jul 2026; sources conflict on the rate.[7]
- April 2026 proclamation sets 50% on core steel, aluminium and copper codes and 25% on derivatives from 6 Apr; the EU contests derivative coverage and no EU tariff-rate quota was confirmed in force.[27]
- New Section 301 tariffs of roughly 10-12.5% on about 60 economies took effect 24 Jul 2026 as Section 122 expired, and USTR said deal caps for the EU and Japan would be upheld.[28]
- On 31 Jul 2026 the Commission removed the 6 Aug expiry and extended indefinitely the suspension of EUR 93bn of countermeasures (Regulation 2025/1564), under continuous review with the option to reactivate; page seen only as a search result.[8]

5. Investment Agenda

US → Germany investment (US capital and firms expanding in Germany)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I12	Batteries components (accumulator parts & separators)	USA	DEU	supply-chain need 0.06 (USA) / 0.68 (DEU)		export credit	medium	2-3 years
I13	Batteries assembly (li-ion cells & packs, other accumulators)	USA	DEU	supply-chain need 0.90 (USA) / 0.88 (DEU)	BMW Group; Tesla	strategic imports	low	2-3 years
I14	Refrigerants	USA	DEU	DEU takes 8% of allied demand (6% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I15	Sensors & controller parts	USA	DEU	DEU takes 6% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I16	AC generators	USA	DEU	DEU takes 16% of allied demand (6% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years
I17	Silver powder	USA	DEU	DEU takes 10% of allied demand (4% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years

Sources: 1.

Germany → US investment (Germany capital and firms expanding in US)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I01	Electric motors	DEU	USA	USA takes 22% of allied demand (15% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I02	Motor & generator parts	DEU	USA	USA takes 18% of allied demand (11% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I03	Power electronics components	DEU	USA	USA takes 18% of allied demand (14% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I04	Oil extraction	DEU	USA	supply-chain need 0.47 (USA) / 0.90 (DEU)		anchor offtake	medium	2-3 years
I05	Electric two-wheelers & works trucks	DEU	USA	USA takes 15% of allied demand (9% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I06	Static converters & inverters	DEU	USA	USA takes 22% of allied demand (17% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I07	Solar thermal water heaters	DEU	USA	USA takes 3% of allied demand (2% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I08	Non-oriented electrical steel (NGO)	DEU	USA	USA takes 0% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I09	Electric furnaces & heating	DEU	USA	USA takes 17% of allied demand (5% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I10	Furnace parts	DEU	USA	USA takes 38% of allied demand (29% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I11	Nuclear reactors & enrichment machinery	DEU	USA	USA takes 8% of allied demand (7% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years

Sources: 1.

Joint investment (at home or in third countries)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I18	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	USA+DEU	third countries (THA; MYS; JPN) and either partner	supply-chain need 0.90 (USA) / 0.95 (DEU)		anchor offtake	high	2-3 years
I19	Synthetic graphite (anode-grade)	USA+DEU	third countries (JPN) and either partner	USA takes 20% of allied demand (14% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I20	Electric & plug-in cars	USA+DEU	third countries (JPN) and either partner	USA takes 18% of allied demand (11% of allied net imports); AIB class BUILD	Mercedes-Benz; Volkswagen; Scout Motors	anchor offtake	medium	2-3 years
I21	Ferrite & other magnets	USA+DEU	third countries (JPN) and either partner	USA takes 8% of allied demand (5% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I22	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	USA+DEU	third countries (CHL; ARG; BRA) and either partner	supply-chain need 0.82 (USA) / 0.85 (DEU)	Arafura Rare Earths	anchor offtake	low	2-3 years

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), US and Germany are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: US brings demand and offtake and coordination across compacts; Germany brings capital and export finance and equipment and manufacturing at scale (AIB strategic roles).

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	US INTEREST	GERMANY INTEREST
X01	Malaysia	Electric Vehicles	Refining	SC-MAGNETS	catch up	direct supply share; direct export share	own secure need (USA)	own secure need (DEU)
X02	Thailand	Electric Vehicles	Refining	SC-MAGNETS	catch up	direct export share	own secure need (USA)	own secure need (DEU)
X03	Thailand	Electric Vehicles	Components	SC-EV; SC-MAGNETS	catch up	direct supply share; direct export share	own preserve need (USA)	own preserve need (DEU)
X04	Indonesia	Electric Vehicles	Components	SC-EV; SC-MAGNETS	preserve	direct supply share; direct export share; adjacent stage share	own preserve need (USA)	own preserve need (DEU)
X05	Viet Nam	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share; direct export share	own preserve need (USA)	own catch up need (DEU)
X06	Indonesia	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share	own preserve need (USA)	own catch up need (DEU)
X07	Philippines	Electric Vehicles	Components	SC-EV; SC-MAGNETS	catch up	direct supply share; direct export share	own preserve need (USA)	own preserve need (DEU)
X08	Turkiye	Electric Vehicles	Components	SC-EV; SC-MAGNETS	catch up	direct supply share; direct export share	own preserve need (USA)	own preserve need (DEU)

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	US INTEREST	GERMANY INTEREST
X09	Malaysia	Electric Vehicles	Components	SC-EV; SC-MAGNETS	catch up	direct export share	own preserve need (USA)	own preserve need (DEU)
X10	Mexico	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share; direct export share	own preserve need (USA)	own catch up need (DEU)

Sources: 1.

- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]

7. Recommended Sectoral Compacts

Inference. 5 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	US ROLE	GERMANY ROLE	BILATERAL MECHANISMS	RECOMMENDED
1	Permanent Magnets & Drive Systems	bilateral priority	4	3	Scale manufacturer	Specialized technology supplier	USDE-M3	TRUE
2	Battery Materials & Cells	bilateral priority	3	2	Resource supplier	Demand anchor	USDE-M3	TRUE
3	Electric Vehicles	bilateral priority	1	2	Scale manufacturer	Scale manufacturer		TRUE
4	Semiconductors & Power Electronics	bilateral priority	1	3	Specialized technology supplier	Scale manufacturer		TRUE
5	Grid Equipment & Conductors	evidence-led candidate	0	2	Demand anchor	Scale manufacturer	USDE-M2	TRUE
6	Heat Pumps & Cooling	secondary	0	2	Processing anchor	Scale manufacturer		FALSE
7	Industrial Electrification & Hydrogen	secondary	0	2	Demand anchor	Scale manufacturer	USDE-M5	FALSE
8	Nuclear Fuel & Reactor Systems	secondary	0	3	Specialized technology supplier	Scale manufacturer		FALSE
9	Solar Manufacturing	secondary	0	1	Demand anchor	Processing anchor		FALSE
10	Wind & Hydropower Equipment	not a bilateral priority	0	0	Demand anchor	Scale manufacturer		FALSE

Sources: 1.

1. Permanent Magnets & Drive Systems

Strategic objective

- AIB mission: Build missing capacity in rare-earth permanent magnets (magnets & core steel)..[1]
- *Recommendation.* Change the outcome at rare-earth oxides & compounds, ferrite & other magnets, electric motors: geopolitical chokepoint; trusted partner capacity constraint.

Why this pair?

- *Inference.* Rare-earth oxides & compounds: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 84% of world supply). Ferrite & other magnets: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Electric motors: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Motor & generator parts: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Rare-earth permanent magnets: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 64% of world supply). Non-oriented electrical steel (NGO): Complementary specialisation (DEU is strong here and the other country is strong at another step of the same chain). AC generators: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Scale manufacturer (lead node ac-generators); Germany Specialized technology supplier (lead node non-oriented-electrical-steel-ngo).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	rare-earth-ores-monazite	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	rare-earth-oxides-and-compounds	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Components	rare-earth-permanent-magnets	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Assembly	electric-motors	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Deployment	Electric Vehicles/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Electric Vehicles/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: supply of ac generators; long-term demand and offtake for ferrite & other magnets, electric motors, motor & generator parts, non-oriented electrical steel (ngo); procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Germany provides: capacity and technology in ferrite & other magnets, electric motors, motor & generator parts, non-oriented electrical steel (ngo); long-term demand for ac generators; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KfW, KfW IPEX, Hermes, UFK, DEG).

- Jointly: pooled offtake and finance for non-coercive rare-earth oxides & compounds, rare-earth permanent magnets.
- Third-country partners needed: JPN, MYS, MEX, PHL, VNM, TWN, KOR.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MYS at rare-earth oxides & compounds; JPN at rare-earth permanent magnets; VNM at rare-earth metals. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for rare-earth oxides & compounds; anchor_offtake for ferrite & other magnets; anchor_offtake for electric motors; anchor_offtake for motor & generator parts; anchor_offtake for rare-earth permanent magnets; standards_alignment for non-oriented electrical steel (ngo); standards_alignment for ac generators

Priority transactions

- *Recommendation.* Long-term offtake by US and Germany buyers for non-coercive rare-earth oxides & compounds, sized to underwrite new capacity (candidate hosts: JPN, MYS) (D02).

Success by 2030

- Rare-earth oxides & compounds: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply).
- Ferrite & other magnets: coercive share of supply to the two buyers below 50% (CHN now holds 58% of world supply); Germany's share of US's imports up from 3%.
- Electric motors: allied residual gap under the China shock closed (now 21% of allied demand); Germany's share of US's imports up from 8%.
- Motor & generator parts: allied residual gap under the China shock closed (now 18% of allied demand); Germany's share of US's imports up from 3%.
- Rare-earth permanent magnets: coercive share of supply to the two buyers below 50% (CHN now holds 64% of world supply).
- Non-oriented electrical steel (NGO): allied residual gap under the China shock closed (now 17% of allied demand); Germany's share of US's imports up from 0%.
- AC generators: allied residual gap under the China shock closed (now 4% of allied demand); US's share of Germany's imports up from 35%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- e-VAC was awarded about US$112m in Section 48C tax credits for its US$500m-plus Sumter facility in April 2024.[14]
- US-listed Energy Fuels agreed on 23 Jun 2026 to buy 100% of Hanau-based VAC from Ara Partners (US$718m cash plus shares); close expected early 2027 subject to US and European approvals; price may adjust with the GM arbitration.[22]

- KfW's Raw Materials Fund (part of the Germany Fund) made its first two equity decisions, EUR 150m in Vulcan Energy (lithium) and EUR 50m in Arafura's Nolans rare earths project; a third is in due diligence and the target size is EUR 1.5bn; no US-linked project named; USD at about 1.16 per EUR.[23]
- German-owned VAC's Sumter plant (about US$500m, nearly 2,000 t/yr, operating since late 2025) shipped first magnets and held an opening ceremony in July 2026; about 90% of output was contracted to GM; expansion room reported up to 12,000 t.[24]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]
- GM terminated its long-term magnet supply agreement with e-VAC (reported 22 Oct 2026); the parties are reportedly in arbitration and the dispute dates from April 2026; one German source dates the notice to July 2026; reasons not stated by the parties.[9]

2. Battery Materials & Cells

Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at synthetic graphite (anode-grade), natural graphite, other accumulators: geopolitical chokepoint; trusted partner capacity constraint.

Why this pair?

- *Inference.* Synthetic graphite (anode-grade): Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Natural graphite: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 82% of world supply). Other accumulators: Shared vulnerability (both need the node and both import heavily from coercive suppliers). Li-ion cells & packs: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 84% of world supply). Battery recycling feed: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Resource supplier (lead node battery-recycling-feed); Germany Demand anchor (lead node li-ion-cells-and-packs).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	natural-graphite	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Refining	synthetic-graphite-anode-grade	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Components	accumulator-parts-and-separators	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Assembly	li-ion-cells-and-packs	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Deployment	Batteries/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Recovery	battery-recycling-feed	AIB node	strong	moderate	complementary specialization	COORDINATE	standards alignment

Sources: 1.

Reciprocal bargain

- US provides: supply of battery recycling feed; long-term demand and offtake for synthetic graphite (anode-grade); procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Germany provides: capacity and technology in synthetic graphite (anode-grade); long-term demand for battery recycling feed; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KfW, KfW IPEX, Hermes, UFK, DEG).
- Jointly: pooled offtake and finance for non-coercive natural graphite, other accumulators, li-ion cells & packs.
- Third-country partners needed: JPN, MDG, MOZ, VNM, POL, HUN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MDG at natural graphite; VNM at other accumulators; POL at li-ion cells & packs; KOR at cathode materials & precursors; FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; KOR at accumulator parts & separators; TWN at battery materials (foil, electrolyte); JPN at manganese dioxide (emd). The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for synthetic graphite (anode-grade); anchor_offtake for natural graphite; anchor_offtake for other accumulators; anchor_offtake for li-ion cells & packs; standards_alignment for battery recycling feed

Priority transactions

- *Recommendation.* Long-term offtake by US and Germany buyers for non-coercive natural graphite, sized to underwrite new capacity (candidate hosts: MDG, MOZ) (D01).

Success by 2030

- Synthetic graphite (anode-grade): coercive share of supply to the two buyers below 50% (CHN now holds 94% of world supply); Germany's share of US's imports up from 7%.
- Natural graphite: coercive share of supply to the two buyers below 50% (CHN now holds 82% of world supply).
- Other accumulators: allied residual gap under the China shock closed (now 18% of allied demand).
- Li-ion cells & packs: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply).
- Battery recycling feed: allied residual gap under the China shock closed (now 8% of allied demand); US's share of Germany's imports up from 3%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- SGL's anode funding found is European (EUR 42.9m IPCEI, Meitingen); the March 2026 call summary reports it closed Lavradio and idled Moses Lake in 2025 (a carbon-fibre site); no 2026 US anode plant or US offtake was found.[29]
- US-owned Tesla raised planned on-site 4680 cell capacity at Gruenheide from 8 to 18 GWh a year with about US$250m, cell output from H1 2027; the plant currently uses CATL and LG Energy Solution cells.[19]
- BMW's US$1.7bn US programme includes a roughly US$700m Woodruff pack plant using Gen6 cylindrical cells from AESC (Florence SC) and iX5 production at Spartanburg, with series pack output reported for Dec 2026; publication date approximate.[21]

3. Electric Vehicles

Strategic objective

- AIB mission: Monitor: existing trade restores supply in every node the model could test (2 nodes the model could not test)..[1]
- *Recommendation.* Change the outcome at electric & plug-in cars, electric two-wheelers & works trucks, electric rail vehicles: geopolitical chokepoint; trusted partner capacity constraint; strategic manufacturing gap.

Why this pair?

- *Inference.* Electric & plug-in cars: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Electric two-wheelers & works trucks: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Electric rail vehicles: Complementary specialisation (DEU is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Scale manufacturer (lead node electric-buses-trucks-and-tractors); Germany Scale manufacturer (lead node electric-and-plug-in-cars).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	Electric Vehicles/Extraction	OSI cell (no AIB node at this stage)	moderate	weak	capacity constrained trusted supply	BUILD	anchor offtake
Refining	Electric Vehicles/Refining	OSI cell (no AIB node at this stage)	moderate	weak	capacity constrained trusted supply	BUILD	anchor offtake
Components	Electric Vehicles/Components	OSI cell (no AIB node at this stage)	strong	strong	direct competition	MANAGE SPECIALIZATION	
Assembly	electric-and-plug-in-cars	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Deployment	Electric Vehicles/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Electric Vehicles/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: long-term demand and offtake for electric & plug-in cars, electric two-wheelers & works trucks, electric rail vehicles; procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Germany provides: capacity and technology in electric & plug-in cars, electric two-wheelers & works trucks, electric rail vehicles; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KfW, KfW IPEX, Hermes, UFK, DEG).
- Jointly: common specifications and qualification.
- Third-country partners needed: JPN, ITA, SWE.

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for electric & plug-in cars; anchor_offtake for electric two-wheelers & works trucks; standards_alignment for electric rail vehicles

Priority transactions

- *Recommendation.* Long-term offtake by US buyers for Germany electric & plug-in cars, sized to underwrite new capacity (candidate hosts: JPN) (D05).

Success by 2030

- Electric & plug-in cars: coercive share of supply to the two buyers below 50% (CHN now holds 69% of world supply); Germany's share of US's imports up from 8%.
- Electric two-wheelers & works trucks: allied residual gap under the China shock closed (now 22% of allied demand); Germany's share of US's imports up from 8%.
- Electric rail vehicles: allied residual gap under the China shock closed (now 0% of allied demand); Germany's share of US's imports up from n/a.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Joint statement caps combined MFN plus Section 232 at 15% for EU pharmaceuticals, semiconductors and lumber, and applies a combined 15% to EU autos and parts (conditional on the EU tabling its tariff legislation); steel and aluminium left to possible tariff-rate quotas.[26]
- Mercedes plans a new Tuscaloosa model by 2027 (GLC localisation reported later); VW's Scout plant in South Carolina (about US$2bn) targets EV SUVs and pickups from late 2026-2027, while ID.4 output at Chattanooga was reported ending April 2026; sources conflict.[18]
- BMW's US$1.7bn US programme includes a roughly US$700m Woodruff pack plant using Gen6 cylindrical cells from AESC (Florence SC) and iX5 production at Spartanburg, with series pack output reported for Dec 2026; publication date approximate.[21]

4. Semiconductors & Power Electronics

Strategic objective

- AIB mission: Monitor: no intervention class reaches the materiality floor (25% of the compact's supply to shift or US$1bn); targeted: build missing capacity in piezo-electric quartz (semiconductor materials); preserve and second-source specialized production in piezo-electric quartz (semiconductor materials)..[1]
- *Recommendation.* Change the outcome at power electronics components, static converters & inverters, sensors & controller parts: trusted partner capacity constraint.

Why this pair?

- *Inference.* Power electronics components: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Static converters & inverters: Capacity-constrained trusted supply (DEU is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Sensors & controller parts: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain). Semiconductor manufacturing equipment: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Specialized technology supplier (lead node semiconductor-manufacturing-equipment); Germany Scale manufacturer (lead node automatic-controllers).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	Electric Grid/Extraction	OSI cell (no AIB node at this stage)	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	piezo-electric-quartz	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Components	power-electronics-components	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Assembly	static-converters-and-inverters	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Deployment	Electric Grid/Deployment	OSI cell (no AIB node at this stage)	not measured	not measured	low strategic case	LOW INTERVENTION	
Recovery	Electric Grid/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: supply of sensors & controller parts, semiconductor manufacturing equipment; long-term demand and offtake for power electronics components, static converters & inverters; procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Germany provides: capacity and technology in power electronics components, static converters & inverters; long-term demand for sensors & controller parts, semiconductor manufacturing equipment; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KfW, KfW IPEX, Hermes, UFK, DEG).
- Jointly: common specifications and qualification.
- Third-country partners needed: JPN.

- Better placed than this pair (largest trusted supplier once China is removed, AIB): PHL at piezo-electric quartz; BRA at gallium, germanium & indium; MYS at power semiconductors. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for power electronics components; anchor_offtake for static converters & inverters; standards_alignment for sensors & controller parts; standards_alignment for semiconductor manufacturing equipment

Priority transactions

- *Recommendation.* Long-term offtake by US buyers for Germany power electronics components, sized to underwrite new capacity (candidate hosts: JPN) (D04).

Success by 2030

- Power electronics components: allied residual gap under the China shock closed (now 31% of allied demand); Germany's share of US's imports up from 5%.
- Static converters & inverters: allied residual gap under the China shock closed (now 33% of allied demand); Germany's share of US's imports up from 5%.
- Sensors & controller parts: allied residual gap under the China shock closed (now 22% of allied demand); US's share of Germany's imports up from n/a.
- Semiconductor manufacturing equipment: allied residual gap under the China shock closed (now 0% of allied demand); US's share of Germany's imports up from 5%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Infineon agreed in Feb 2025 to sell its 200 mm Austin fab to SkyWater with a long-term supply agreement, reducing its own US wafer capacity (about 4,000 US staff remain).[15]

5. Grid Equipment & Conductors (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Germany world share: copper smelting & refining 1.3%, copper wire & rod 11.1%, grain-oriented electrical steel (goes) 8.3%, insulators & bushings 8.7%, electricity meters 1.2%, transformers 4.0%, switchgear & protection 11.2%, copper ores & concentrates 0.0%, overhead conductors 0.4%, power cables 4.4%, aluminium wire rod 0.4%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: copper smelting & refining (US high 0.90, 15% of allied demand; Germany high 0.78); copper wire & rod (US high 0.96, 10% of allied demand; Germany low (cell 0.78; not a material buyer)); insulators & bushings (US demand 0.32, 19% of allied demand; Germany low 0.33); electricity meters (US demand 0.33, 22% of allied demand; Germany demand 0.14); transformers (US demand 0.33, 40% of allied demand; Germany demand 0.14); switchgear & protection (US demand 0.33, 20% of allied demand; Germany low 0.14); copper ores & concentrates (US low (cell 0.89; not a material buyer), 0% of allied demand; Germany high 0.94); overhead conductors (US demand 0.33, 32% of allied demand; Germany demand 0.14);

power cables (US demand 0.33, 20% of allied demand; Germany demand 0.14); aluminium wire rod (US demand n/a, 24% of allied demand; Germany demand n/a).[1]

Strategic objective

- AIB mission: Monitor: no intervention class reaches the materiality floor (25% of the compact's supply to shift or US$1bn); targeted: build missing capacity in grain-oriented electrical steel (GOES) (wire rod & transformer steel)..[1]
- *Recommendation.* Change the outcome at copper smelting & refining, copper wire & rod: geopolitical chokepoint; no material intervention.

Why this pair?

- *Inference.* Copper smelting & refining: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 50% of world supply). Copper wire & rod: Germany supplies / US needs (DEU is a capable trusted supplier and the other side needs the node).
- AIB compact roles: US Demand anchor (lead node transformers); Germany Scale manufacturer (lead node switchgear-and-protection).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	GERMANY POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	copper-ores-and-concentrates	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	copper-smelting-and-refining	AIB node	weak	weak	shared vulnerability	BUILD	anchor offtake
Components	insulators-and-bushings	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Assembly	electricity-meters	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Electric Grid/Deployment	OSI cell (no AIB node at this stage)	not measured	not measured	low strategic case	LOW INTERVENTION	
Recovery	Electric Grid/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: long-term demand and offtake for copper wire & rod; procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Germany provides: capacity and technology in copper wire & rod; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KfW, KfW IPEX, Hermes, UFK, DEG).
- Jointly: pooled offtake and finance for non-coercive copper smelting & refining.
- Third-country partners needed: none identified by the AIB host model.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): JPN at grain-oriented electrical steel (goes); ITA at insulators & bushings; MEX at electricity meters. The bilateral compact should not

try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for copper smelting & refining; NA for copper wire & rod

Priority transactions

- *Recommendation.* Long-term offtake by US and Germany buyers for non-coercive copper smelting & refining, sized to underwrite new capacity (D07).

Success by 2030

- Copper smelting & refining: coercive share of supply to the two buyers below 50% (CHN now holds 50% of world supply).
- Copper wire & rod: allied residual gap under the China shock closed (now 0% of allied demand); Germany's share of US's imports up from 0%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Siemens Energy announced a US$150m large power transformer plant in Charlotte in Feb 2024, originally for early-2026 output (24 units a year, rising to 57); a June 2025 report put US production at 2027.[13]
- Siemens announced more than US$10bn of added US investment in March 2025, including electrical-products plants in Fort Worth and Pomona (US$285m reported) and the Altair acquisition; the figure combines capex and an acquisition.[16]
- Siemens Energy announced US$1bn of US expansion across transformers, grid equipment and turbines, with a greenfield high-voltage switchgear plant in Mississippi (up to 300 jobs).[17]
- April 2026 proclamation sets 50% on core steel, aluminium and copper codes and 25% on derivatives from 6 Apr; the EU contests derivative coverage and no EU tariff-rate quota was confirmed in force.[27]
- the US$800m figure is from a secondary source; publication date approximate.[20]

8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.
- *Inference.* Result: **upgrade**; coverage 0% of recommended compacts; cross-sector mechanism with KPIs and review: no.

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-MAGNETS	TRUE	FALSE	FALSE
SC-BATTERIES	TRUE	FALSE	FALSE
SC-EV	FALSE	FALSE	FALSE

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-SEMIS	FALSE	FALSE	FALSE
SC-GRID	FALSE	FALSE	FALSE

Sources: 1.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
USDE-M1	US-EU framework trade arrangement and its 2026 legal basis (EU-level, applies to Germany)	in force	cross sector	FALSE	partial	FALSE	partial	TRUE
USDE-M2	EU countermeasures and Section 232 friction (EU-level, applies to Germany)	suspended	SC-GRID/cross sector	FALSE	FALSE	partial	partial	FALSE
USDE-M3	US-EU critical minerals MOU and action plan (EU-level, applies to Germany)	signed	SC-MAGNETS/SC-BATTERIES/cross sector	FALSE	FALSE	FALSE	partial	TRUE
USDE-M4	G7 Critical Minerals Action Plan and Minerals Security Partnership	operating	cross sector	partial	FALSE	FALSE	partial	FALSE
USDE-M5	US-Germany Climate and Energy Partnership (2022) and TTC	unknown	SC-INDUSTRY/cross sector	FALSE	FALSE	FALSE	partial	FALSE

Sources: 1.

Recommended: upgrade US-EU critical minerals MOU and action plan (EU-level, applies to Germany) into the execution framework for the compacts

- **Scope:** Permanent Magnets & Drive Systems; Battery Materials & Cells; Electric Vehicles; Semiconductors & Power Electronics; Grid Equipment & Conductors; no other sectors.
- **Reciprocal commitments:** US commits demand certainty (procurement recognition and anchor offtake) and market-access predictability for the nodes Germany supplies (synthetic graphite (anode-grade), electric & plug-in cars, ferrite & other magnets, electric motors); Germany commits the same for the nodes US supplies (battery recycling feed, sensors & controller parts, ac generators, semiconductor manufacturing equipment). Each commits public finance for the compact nodes, at home, in the partner or with named third-country hosts.
- **Compact portfolio:** Permanent Magnets & Drive Systems (bilateral priority); Battery Materials & Cells (bilateral priority); Electric Vehicles (bilateral priority); Semiconductors & Power Electronics (bilateral priority); Grid Equipment & Conductors (evidence-led candidate).
- **Financing lanes:** US: DOE LPO, DoW OSC, DFC, EXIM; Germany: KfW, KfW IPEX, Hermes, UFK, DEG; demand-side tools (offtake, stockpiles, procurement): DOE, DoW OSC, DLA and Hermes, UFK, DERA, BAFA; third-country finance: DFC and DEG, BMZ.
- **Governance:** one bilateral steering line inside US-EU critical minerals MOU and action plan (EU-level, applies to Germany). Projects are selected by a joint committee, not by one side.
- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle () unless renewed on evidence.

9. 12–36 Month Deal Pipeline

Inference. Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per compact.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
1	Long-term offtake by US and Germany buyers for non-coercive natural graphite, sized to underwrite new capacity (candidate hosts: MDG, MOZ)	SC-BATTERIES	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	high	high
2	Long-term offtake by US and Germany buyers for non-coercive rare-earth oxides & compounds, sized to underwrite new capacity (candidate hosts: JPN, MYS)	SC-MAGNETS	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	high	high
3	Long-term offtake by US and Germany buyers for non-coercive nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)), sized to underwrite new capacity	SC-NUCLEAR	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	medium	high
4	Long-term offtake by US buyers for Germany power electronics components, sized to underwrite new capacity (candidate hosts: JPN)	SC-SEMIS	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	low	high
5	Long-term offtake by US buyers for Germany electric & plug-in cars, sized to underwrite new capacity (candidate hosts: JPN)	SC-EV	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	high	medium
6	Long-term offtake by Germany buyers for US oil extraction, sized to underwrite new capacity		anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	low	medium
7	Long-term offtake by US and Germany buyers for non-coercive copper smelting & refining, sized to underwrite new capacity	SC-GRID	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	high	medium	low
8	Long-term offtake by US buyers for Germany furnace parts, sized to underwrite new capacity (candidate hosts: ITA, JPN)	SC-INDUSTRY	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	medium	low	high
9	Long-term offtake by Germany buyers for US refrigerants, sized to underwrite new capacity (candidate hosts: NLD)	SC-HEAT	anchor offtake	USA: DOE, DoW OSC, DLA; DEU: DERA	Signed long-term offtake from at least one large buyer	medium	low	high

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
10	Joint specification work plan for nuclear reactors & enrichment machinery	SC-NUCLEAR	standards alignment	USA: USTR, NRC; DEU: BMWE	Joint list of specifications that fragment the market and a work plan to align them	medium	low	medium
11	Joint specification work plan for silver powder	SC-SOLAR	standards alignment	USA: USTR, NRC; DEU: BMWE	Joint list of specifications that fragment the market and a work plan to align them	medium	low	medium

Sources: 1.

10. What Not to Do

- *Recommendation.* **N01** Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity: both are strong suppliers (US 7%, Germany 8%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N02** Do not co-subsidise new Automatic controllers capacity: both are strong suppliers (US 16%, Germany 12%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N03** Do not co-subsidise new Electric buses, trucks & tractors capacity: both are strong suppliers (US 15%, Germany 11%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* **N04** Do not create a new bilateral energy MOU or a free-standing MEA institution: USDE-M3 already exists (a standing framework with no pipeline or finance yet); dock the compacts in it.
- *Recommendation.* **N05** Do not put new public money behind Batteries assembly (li-ion cells & packs, other accumulators) investment that is already committed (evidence USDE-23; USDE-25); support is additional only beyond it.
- *Recommendation.* **N06** Do not put new public money behind Copper smelting & refining investment that is already committed (evidence USDE-22); support is additional only beyond it.

11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 256 items (partial 197, unverified 18, verified 41).[1]

Where the findings are weak or counterintuitive

- Joint bets resting on a node need inherited from its cell (a proxy): Synthetic graphite (anode-grade); Electric & plug-in cars; Rare-earth oxides & compounds; Power electronics components.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): Grid Equipment & Conductors.
- Germany has no listed trade position (under 1% of world supply) at nodes where the US has material demand: rare-earth oxides & compounds, piezo-electric quartz, blade core wood (balsa), uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), overhead conductors, aluminium wire rod, direct-reduced iron (dri/hbi). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.

- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 (oil extraction; buyer: Germany): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

Sources

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Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

Table A1. Section 2, What Each Country Is Good At — and What It Needs

COMPACT	NODE	STAGE	US SHARE	US NEED	GERMANY SHARE	GERMANY NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-BATTERIES	Natural graphite	Extraction	3.1%	high 0.82	6.4%	high 0.85	BUILD	CHN 82%
SC-BATTERIES	Battery recycling feed	Recovery	43.7%	low (cell 0.88; not a material buyer)	8.0%	low (cell 0.92; not a material buyer)	LOW INTERVENTION	CHN 8%
SC-BATTERIES	Lithium minerals	Extraction	4.0%	low (cell 0.82; not a material buyer)	1.6%	low (cell 0.85; not a material buyer)	LOW INTERVENTION	CHN 25%
SC-BATTERIES	Cobalt intermediates & chemicals	Refining	2.4%	demand 0.45	2.2%	demand 0.35	BUILD	CHN 76%
SC-BATTERIES	Lithium chemicals	Refining	3.1%	demand 0.45	1.1%	demand 0.35	BUILD	CHN 74%
SC-BATTERIES	Manganese dioxide (EMD)	Refining	3.7%	demand 0.45	1.6%	demand 0.35	BUILD	CHN 37%
SC-BATTERIES	Synthetic graphite (anode-grade)	Refining	5.7%	demand 0.45	6.2%	demand 0.35	BUILD	CHN 94%
SC-BATTERIES	Cathode materials & precursors	Refining	4.9%	low 0.45	2.9%	low 0.35	BUILD	CHN 92%
SC-BATTERIES	Accumulator parts & separators	Components	3.1%	high 0.73	4.0%	high 0.83	BUILD	CHN 38%
SC-BATTERIES	Battery materials (foil, electrolyte)	Refining	2.8%	demand 0.45	3.0%	low 0.35	BUILD	CHN 16%
SC-BATTERIES	Other accumulators	Assembly	7.4%	high 0.90	7.0%	high 0.88	BUILD	CHN 28%
SC-BATTERIES	Li-ion cells & packs	Assembly	6.4%	high 0.90	3.1%	high 0.88	BUILD	CHN 84%
SC-EV	Electric & plug-in cars	Assembly	7.1%	demand 0.15	25.6%	low 0.01	BUILD	CHN 69%
SC-EV	Electric two-wheelers & works trucks	Assembly	5.4%	demand 0.15	20.1%	low 0.01	BUILD	CHN 30%
SC-EV	Electric rail vehicles	Assembly	0.5%	low n/a	20.7%	low n/a	LOW INTERVENTION	CHN 14%
SC-EV	Electric buses, trucks & tractors	Assembly	14.6%	low 0.15	10.6%	low 0.01	BUILD	CHN 24%
SC-GRID	Copper ores & concentrates	Extraction	4.7%	low (cell 0.89; not a material buyer)	0.0%	high 0.94	LOW INTERVENTION	CHN 13%
SC-GRID	Copper smelting & refining	Refining	1.4%	high 0.90	1.3%	high 0.78	LOW INTERVENTION	CHN 50%
SC-GRID	Copper wire & rod	Refining	4.9%	high 0.96	11.1%	low (cell 0.78; not a material buyer)	LOW INTERVENTION	CHN 6%
SC-GRID	Grain-oriented electrical steel (GOES)	Refining	3.7%	low (cell 0.90; not a material buyer)	8.3%	low (cell 0.78; not a material buyer)	BUILD	CHN 41%

COMPACT	NODE	STAGE	US SHARE	US NEED	GERMANY SHARE	GERMANY NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-GRID	Aluminium wire rod	Refining	0.1%	demand n/a	0.4%	demand n/a	LOW INTERVENTION	RUS 7%
SC-GRID	Insulators & bushings	Components	6.6%	demand 0.32	8.7%	low 0.33	BUILD	CHN 32%
SC-GRID	Electricity meters	Assembly	6.3%	demand 0.33	1.2%	demand 0.14	BUILD	CHN 34%
SC-GRID	Transformers	Assembly	2.0%	demand 0.33	4.0%	demand 0.14	LOW INTERVENTION	CHN 21%
SC-GRID	Switchgear & protection	Assembly	5.4%	demand 0.33	11.2%	low 0.14	LOW INTERVENTION	CHN 20%
SC-GRID	Overhead conductors	Components	1.0%	demand 0.33	0.4%	demand 0.14	LOW INTERVENTION	CHN 24%
SC-GRID	Power cables	Components	6.7%	demand 0.33	4.4%	demand 0.14	LOW INTERVENTION	CHN 19%
SC-HEAT	Refrigerants	Refining	19.9%	low n/a	1.6%	demand n/a	BUILD	CHN 45%
SC-HEAT	Refrigeration & heat-pump compressors	Components	7.6%	demand n/a	6.8%	low n/a	BUILD	CHN 33%
SC-HEAT	Solar thermal water heaters	Assembly	0.4%	demand n/a	5.8%	demand n/a	BUILD	CHN 45%
SC-HEAT	Heat pumps & AC units	Assembly	4.0%	demand n/a	3.0%	demand n/a	BUILD	CHN 49%
SC-INDUSTRY	PGMs (iridium, ruthenium)	Refining	4.0%	demand n/a	15.9%	low n/a	LOW INTERVENTION	RUS 30%
SC-INDUSTRY	Electrolysers	Assembly	7.5%	demand n/a	7.4%	low n/a	BUILD	CHN 32%
SC-INDUSTRY	Hydrogen, ammonia & methanol	Refining	8.7%	low n/a	2.1%	demand n/a	LOW INTERVENTION	RUS 3%
SC-INDUSTRY	Furnace parts	Components	7.7%	demand n/a	13.9%	low n/a	BUILD	CHN 21%
SC-INDUSTRY	Graphite electrodes	Components	2.3%	demand n/a	3.2%	low n/a	COORDINATE	CHN 38%
SC-INDUSTRY	Direct-reduced iron (DRI/HBI)	Refining	5.2%	demand n/a	0.0%	demand n/a	LOW INTERVENTION	RUS 20%
SC-INDUSTRY	Electric furnaces & heating	Assembly	8.0%	demand n/a	19.4%	low n/a	BUILD	CHN 27%
SC-MAGNETS	Rare-earth ores (monazite)	Extraction	9.2%	low (cell 0.82; not a material buyer)	0.0%	low (cell 0.85; not a material buyer)	INSUFFICIENT DATA	CHN 59%
SC-MAGNETS	Rare-earth oxides & compounds	Refining	5.3%	high 0.96	1.0%	high 0.95	BUILD	CHN 84%
SC-MAGNETS	Rare-earth metals	Refining	5.3%	low (cell 0.96; not a material buyer)	0.0%	low (cell 0.95; not a material buyer)	BUILD	CHN 84%
SC-MAGNETS	Ferrite & other magnets	Components	2.8%	high 0.73	6.2%	high 0.83	BUILD	CHN 58%
SC-MAGNETS	Rare-earth permanent magnets	Components	1.8%	high 0.73	4.0%	high 0.83	BUILD	CHN 64%
SC-MAGNETS	Non-oriented electrical steel (NGO)	Refining	1.4%	low (cell 0.90; not a material buyer)	16.5%	low (cell 0.78; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	Motor & generator parts	Components	6.5%	high 0.73	13.6%	low (cell 0.83; not a material buyer)	BUILD	CHN 26%

COMPACT	NODE	STAGE	US SHARE	US NEED	GERMANY SHARE	GERMANY NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-MAGNETS	Electric motors	Assembly	5.4%	high 0.73	11.4%	low (cell 0.83; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	AC generators	Assembly	27.4%	low 0.27	8.5%	demand 0.32	LOW INTERVENTION	CHN 15%
SC-NUCLEAR	Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores)	Extraction	0.7%	high 0.99	0.0%	low (cell 0.96; not a material buyer)	INSUFFICIENT DATA	RUS 8%
SC-NUCLEAR	Uranium conversion & enrichment	Refining	3.3%	high 0.99	5.4%	demand 0.61	LOW INTERVENTION	RUS 16%
SC-NUCLEAR	Fuel assemblies & zirconium	Components	9.2%	low 0.48	10.9%	low 0.48	LOW INTERVENTION	RUS 28%
SC-NUCLEAR	Nuclear special materials	Refining	3.5%	demand n/a	2.4%	demand n/a	LOW INTERVENTION	0%
SC-NUCLEAR	Reactor parts	Components	18.1%	low 0.48	2.6%	low 0.48	LOW INTERVENTION	RUS 45%
SC-NUCLEAR	Nuclear reactors & enrichment machinery	Assembly	0.7%	demand 0.47	25.5%	low 0.06	LOW INTERVENTION	RUS 49%
SC-SEMIS	Piezo-electric quartz	Refining	3.3%	demand n/a	0.7%	low n/a	BUILD	CHN 60%
SC-SEMIS	Gallium, germanium & indium	Refining	6.5%	demand n/a	5.2%	demand n/a	BUILD	CHN 25%
SC-SEMIS	Semiconductor manufacturing equipment	Assembly	15.6%	low 0.20	7.5%	low 0.14	LOW INTERVENTION	CHN 5%
SC-SEMIS	Sensors & controller parts	Components	10.1%	low n/a	9.9%	low n/a	BUILD	CHN 24%
SC-SEMIS	Power semiconductors	Components	2.5%	demand 0.32	7.0%	demand 0.33	BUILD	CHN 32%
SC-SEMIS	Integrated circuits	Components	3.8%	low n/a	1.3%	low n/a	LOW INTERVENTION	CHN 16%
SC-SEMIS	Power electronics components	Components	2.8%	high 0.73	6.8%	high 0.83	BUILD	CHN 37%
SC-SEMIS	Static converters & inverters	Assembly	4.3%	demand 0.28	7.2%	demand 0.14	BUILD	CHN 38%
SC-SEMIS	Automatic controllers	Assembly	15.6%	low 0.28	12.4%	low 0.12	LOW INTERVENTION	CHN 12%
SC-SOLAR	Polysilicon	Refining	30.7%	low (cell 0.71; not a material buyer)	33.8%	low 0.47	LOW INTERVENTION	CHN 97%
SC-SOLAR	Silver powder	Refining	25.8%	low n/a	2.2%	demand n/a	LOW INTERVENTION	0%
SC-SOLAR	Wafers	Refining	9.9%	low 0.21	5.8%	low 0.19	BUILD	CHN 98%
SC-SOLAR	Solar cells	Components	0.9%	demand 0.21	2.0%	low 0.19	BUILD	CHN 92%
SC-SOLAR	PV cover glass	Components	2.5%	demand 0.21	3.0%	low 0.19	BUILD	CHN 61%
SC-SOLAR	PV modules & generators	Assembly	4.5%	demand 0.20	2.0%	demand 0.14	BUILD	CHN 84%
SC-WIND	Blade core wood (balsa)	Refining	0.1%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	CHN 3%
SC-WIND	Wind blades & hubs	Components	8.2%	demand 0.27	8.4%	demand 0.32	BUILD	CHN 71%
SC-WIND	Hydro turbine parts	Components	2.0%	demand n/a	7.9%	low n/a	COORDINATE	CHN 35%
SC-WIND	Wind towers	Components	0.4%	demand 0.27	15.9%	low 0.32	LOW INTERVENTION	CHN 18%

COMPACT	NODE	STAGE	US SHARE	US NEED	GERMANY SHARE	GERMANY NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-WIND	Wind turbine generating sets	Assembly	7.3%	low 0.33	25.3%	low 0.10	BUILD	CHN 74%
SC-WIND	Hydro turbines	Assembly	3.1%	low n/a	33.9%	low n/a	LOW INTERVENTION	CHN 21%

Sources: 1.

Table A2. Section 3, Bilateral Complementarity Map

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	GERMANY POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
1	joint bet	Electric Vehicles	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	Refining	moderate	weak	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.90 / 0.95
2	joint bet	Battery Materials & Cells	Synthetic graphite (anode-grade)	Refining	moderate (5.7% of world supply)	strong (6.2% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.45 / 0.35
3	joint bet	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds	Refining	moderate (5.3% of world supply)	moderate (1.0% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.96 / 0.95
4	joint bet	Battery Materials & Cells	Natural graphite	Extraction	moderate (3.1% of world supply)	moderate (6.4% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.82 / 0.85
5	joint bet	Batteries	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.81 / 0.85
6	joint bet	Electric Vehicles	Electric & plug-in cars	Assembly	moderate (7.1% of world supply)	strong (25.6% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.15 / 0.01
7	joint bet	Battery Materials & Cells	Other accumulators	Assembly	moderate (7.4% of world supply)	moderate (7.0% of world supply)	Shared vulnerability	trusted partner capacity constraint; need 0.90 / 0.88
8	joint bet	Batteries	Batteries components (accumulator parts & separators)	Components	strong	moderate	US supplies / Germany needs	strategic manufacturing gap; need 0.06 / 0.68
9	joint bet	Permanent Magnets & Drive Systems	Ferrite & other magnets	Components	moderate (2.8% of world supply)	strong (6.2% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.73 / 0.83
10	joint bet	Permanent Magnets & Drive Systems	Electric motors	Assembly	moderate (5.4% of world supply)	strong (11.4% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.73 / 0.83
11	joint bet	Permanent Magnets & Drive Systems	Motor & generator parts	Components	moderate (6.5% of world supply)	strong (13.6% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.73 / 0.83
12	joint bet	Nuclear	Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores))	Extraction	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.99 / 0.96
13	joint bet	Semiconductors & Power Electronics	Power electronics components	Components	moderate (2.8% of world supply)	strong (6.8% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.73 / 0.83
14	joint bet	Oil	Oil extraction	Extraction	strong	weak	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.47 / 0.90
15	coordinate	Battery Materials & Cells	Li-ion cells & packs	Assembly	moderate (6.4% of world supply)	moderate (3.1% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.90 / 0.88
16	coordinate	Electric Vehicles	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	moderate	weak	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.82 / 0.85
17	coordinate	Battery Materials & Cells	Battery recycling feed	Recovery	strong (43.7% of world supply)	moderate (8.0% of world supply)	Complementary specialisation	emerging bottleneck; need 0.88 / 0.92
18	coordinate	Batteries	Batteries assembly (li-ion cells & packs, other accumulators)	Assembly	strong	moderate	US supplies / Germany needs	geopolitical chokepoint; need 0.90 / 0.88
19	coordinate	Electric Vehicles	Electric two-wheelers & works trucks	Assembly	moderate (5.4% of world supply)	strong (20.1% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.15 / 0.01

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	GERMANY POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
20	coordinate	Permanent Magnets & Drive Systems	Rare-earth permanent magnets	Components	moderate (1.8% of world supply)	moderate (4.0% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.73 / 0.83
21	coordinate	Solar	Solar components (insulators & bushings, power electronics components, power semiconductors)	Components	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.21 / 0.19
22	coordinate	Gas	Gas extraction	Extraction	strong	weak	US supplies / Germany needs	no material intervention; need 0.30 / 0.99
23	coordinate	Electric Grid	Electric Grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes))	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.90 / 0.78
24	coordinate	Grid Equipment & Conductors	Copper smelting & refining	Refining	weak (1.4% of world supply)	weak (1.3% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.90 / 0.78
25	coordinate	Heat Pumps & Cooling	Refrigerants	Refining	strong (19.9% of world supply)	weak (1.6% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
26	coordinate	Semiconductors & Power Electronics	Static converters & inverters	Assembly	moderate (4.3% of world supply)	strong (7.2% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.28 / 0.14
27	coordinate	Heat Pumps & Cooling	Solar thermal water heaters	Assembly	weak (0.4% of world supply)	moderate (5.8% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
28	coordinate	Semiconductors & Power Electronics	Sensors & controller parts	Components	strong (10.1% of world supply)	moderate (9.9% of world supply)	Complementary specialisation	trusted partner capacity constraint; need n/a / n/a
29	coordinate	Permanent Magnets & Drive Systems	Non-oriented electrical steel (NGO)	Refining	moderate (1.4% of world supply)	strong (16.5% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.90 / 0.78
30	coordinate	Industrial Electrification & Hydrogen	Electric furnaces & heating	Assembly	moderate (8.0% of world supply)	strong (19.4% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
31	coordinate	Industrial Electrification & Hydrogen	Furnace parts	Components	moderate (7.7% of world supply)	strong (13.9% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
32	coordinate	Permanent Magnets & Drive Systems	AC generators	Assembly	strong (27.4% of world supply)	moderate (8.5% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.27 / 0.32
33	coordinate	Nuclear Fuel & Reactor Systems	Nuclear reactors & enrichment machinery	Assembly	weak (0.7% of world supply)	strong (25.5% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.47 / 0.06
34	coordinate	Solar Manufacturing	Silver powder	Refining	strong (25.8% of world supply)	moderate (2.2% of world supply)	Complementary specialisation	strategic manufacturing gap; need n/a / n/a
35	coordinate	Nuclear Fuel & Reactor Systems	Fuel assemblies & zirconium	Components	moderate (9.2% of world supply)	strong (10.9% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.48 / 0.48
36	coordinate	Nuclear Fuel & Reactor Systems	Reactor parts	Components	strong (18.1% of world supply)	moderate (2.6% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.48 / 0.48
37	coordinate	Semiconductors & Power Electronics	Semiconductor manufacturing equipment	Assembly	strong (15.6% of world supply)	moderate (7.5% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.20 / 0.14
38	coordinate	Oil	Oil assembly (ac generators)	Assembly	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.14 / 0.18
39	coordinate	Coal	Coal components	Components	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.49 / 0.50
40	coordinate	Coal	Coal assembly	Assembly	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.42 / 0.39
41	coordinate	Oil	Oil refining	Refining	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.00 / 0.37
42	coordinate	Electric Vehicles	Electric rail vehicles	Assembly	weak (0.5% of world supply)	strong (20.7% of world supply)	Complementary specialisation	strategic manufacturing gap; need n/a / n/a

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	GERMANY POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
43	coordinate	Grid Equipment & Conductors	Copper wire & rod	Refining	moderate (4.9% of world supply)	strong (11.1% of world supply)	Germany supplies / US needs	no material intervention; need 0.96 / 0.78
44	coordinate	Coal	Coal extraction	Extraction	strong	weak	US supplies / Germany needs	no material intervention; need 0.61 / 0.84
45	coordinate	Wind	Wind components (ac generators, ferrite & other magnets, insulators & bushings)	Components	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.27 / 0.32
46	manage competition	Solar Manufacturing	Polysilicon	Refining	strong (30.7% of world supply)	strong (33.8% of world supply)	Direct competition	geopolitical chokepoint; need 0.71 / 0.47
47	manage competition	Electric Vehicles	Electric buses, trucks & tractors	Assembly	strong (14.6% of world supply)	strong (10.6% of world supply)	Direct competition	trusted partner capacity constraint; need 0.15 / 0.01
48	manage competition	Electric Vehicles	Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets)	Components	strong	strong	Direct competition	no material intervention; need 0.73 / 0.83
49	manage competition	Electric Vehicles	Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks)	Assembly	strong	strong	Direct competition	no material intervention; need 0.15 / 0.01
50	manage competition	Solar	Solar assembly (electricity meters, overhead conductors, power cables)	Assembly	strong	strong	Direct competition	no material intervention; need 0.20 / 0.14
51	manage competition	Semiconductors & Power Electronics	Automatic controllers	Assembly	strong (15.6% of world supply)	strong (12.4% of world supply)	Direct competition	no material intervention; need 0.28 / 0.12
52	manage competition	Wind	Wind assembly (electricity meters, overhead conductors, power cables)	Assembly	strong	strong	Direct competition	no material intervention; need 0.33 / 0.10

Sources: 1.

Table A3. Section 4, Trade Agenda

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T01	USA -> DEU	Electric Vehicles/Refining	procurement recognition	Recognise US output of Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds) as trusted domestic-equivalent supply in Germany procurement and content rules; capacity, not access, is the constraint.
T02	DEU -> USA	synthetic-graphite-anode-grade	tariff risk	Bind market access for Germany Synthetic graphite (anode-grade) into US: US's protection priority on this node is in its top third (0.91), which taxes the trusted supplier it needs.
T03	both	rare-earth-oxides-and-compounds	common standards	Common specifications for Rare-earth oxides & compounds so a joint supplier base can serve both markets.
T04	both	natural-graphite	common standards	Common specifications for Natural graphite so a joint supplier base can serve both markets.
T05	both	Batteries/Extraction	common standards	Common specifications for Batteries extraction (cobalt ores, lithium minerals, natural graphite) so a joint supplier base can serve both markets.
T06	DEU -> USA	electric-and-plug-in-cars	procurement recognition	Recognise Germany output of Electric & plug-in cars as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T07	both	other-accumulators	common standards	Common specifications for Other accumulators so a joint supplier base can serve both markets.
T08	USA -> DEU	Batteries/Components	predictability	Keep US exports of Batteries components (accumulator parts & separators) to Germany tariff-free and predictable; the buyer's need is met by the trusted supplier.
T09	DEU -> USA	ferrite-and-other-magnets	procurement recognition	Recognise Germany output of Ferrite & other magnets as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T10	DEU -> USA	electric-motors	procurement recognition	Recognise Germany output of Electric motors as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T11	DEU -> USA	motor-and-generator-parts	procurement recognition	Recognise Germany output of Motor & generator parts as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T12	both	Nuclear/Extraction	common standards	Common specifications for Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)) so a joint supplier base can serve both markets.
T13	open	uranium-fuel-upstream--aggregated	open sourcing	Leave Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores) open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T14	open	nuclear-special-materials	open sourcing	Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T15	open	transformers	open sourcing	Leave Transformers open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T16	open	uranium-conversion-and-enrichment	open sourcing	Leave Uranium conversion & enrichment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T17	open	overhead-conductors	open sourcing	Leave Overhead conductors open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T18	open	wind-towers	open sourcing	Leave Wind towers open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T19	both	polysilicon	no preference	No bilateral preference for Polysilicon: both are strong suppliers; keep the market contestable.
T20	both	electric-buses-trucks-and-tractors	no preference	No bilateral preference for Electric buses, trucks & tractors: both are strong suppliers; keep the market contestable.
T21	both	Electric Vehicles/Components	no preference	No bilateral preference for Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets): both are strong suppliers; keep the market contestable.
T22	both	Electric Vehicles/Assembly	no preference	No bilateral preference for Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks): both are strong suppliers; keep the market contestable.
T23	both	Solar/Assembly	no preference	No bilateral preference for Solar assembly (electricity meters, overhead conductors, power cables): both are strong suppliers; keep the market contestable.
T24	both	automatic-controllers	no preference	No bilateral preference for Automatic controllers: both are strong suppliers; keep the market contestable.
T25	both	Wind/Assembly	no preference	No bilateral preference for Wind assembly (electricity meters, overhead conductors, power cables): both are strong suppliers; keep the market contestable.

Sources: 1.

Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.

Allied Industrial Base Explorer, Partnership memo: United States–Germany Strategic Energy & Industrial Partnership